Reinforcement Learning and Decision Making Under Uncertainty **Spring 2026**

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Project Report

Risk-Sensitive Reinforcement Learning for Study Scheduling: A CVaR-DQN Approach

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Code repository: `https://github.com/muhammadov-q/cvar-dqn-study-scheduling`

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Abstract

We address optimal exam study scheduling using reinforcement learning. A student must allocate limited study days across multiple subjects, balancing knowledge acquisition against forgetting, energy constraints, and subject dependencies. Standard DQN maximizes expected total score but may sacrifice individual exams. We apply **CVaR-DQN**, a distributional RL method that optimizes the Conditional Value at Risk, to produce risk-sensitive schedules that protect against worst-case outcomes. We evaluate seven methods across three environment configurations: CVaR-DQN consistently improves $\text{CVaR}_{10\%}$ over DQN while sacrificing a small amount of mean return, exactly the trade-off the algorithm is designed to make. Ablation studies show energy stochasticity is the primary source of outcome variance, and removing it eliminates the mean–risk gap entirely. As a pilot, we explore *Subject-Constrained CVaR-DQN*—a Multi-Objective Distributional Constrained MDP with per-subject CVaR constraints—which reduces Physics failure on Medium from 30% to 8% but underperforms on aggregate metrics and under distribution shift, motivating adaptive thresholding as future work.

1 Introduction

Students face a sequential decision problem when preparing for exams: which subject to study each day, given limited time, stochastic cognitive energy, and exponential forgetting. Standard reinforcement-learning agents maximize expected total score, which can lead to policies that sacrifice low-weight exams to inflate the aggregate. In practice, failing even one exam carries severe consequences. We apply **CVaR-DQN**—a distributional RL method that optimizes the Conditional Value at Risk [Chow et al., 2015, Dabney et al., 2018]—to produce risk-sensitive study schedules that protect against worst-case outcomes, and additionally explore an extension that imposes per-subject constraints.[1]

Related work. RL has been applied to flashcard review timing [Reddy et al., 2017] and broader educational sequencing [Doroudi et al., 2019]. We build on DQN [Mnih et al., 2015] and Double DQN [Van Hasselt et al., 2016], and extend QR-DQN [Dabney et al., 2018] with CVaR action selection [Chow et al., 2015]. Our pilot extension combines CMDP-style constrained optimization [Altman, 1999, Achiam et al., 2017] with distributional RL by constraining per-subject CVaR.

Contributions. (i) A Gymnasium environment modeling exam scheduling with forgetting, energy, and prerequisites; (ii) a comparison of seven scheduling methods across three environment scales with ablation and robustness analyses; (iii) empirical evidence that CVaR-DQN consistently improves worst-case $\text{CVaR}_{10\%}$ over DQN at a small cost in mean return; (iv) a pilot study of *Subject-Constrained CVaR-DQN*, a Multi-Objective Distributional Constrained MDP formulation, with positive and negative findings.

2 Preliminaries

We model exam preparation as a finite-horizon MDP. The **state** is $s = (\mathbf{k}, e, d)$ where $\mathbf{k} \in \{0, \ldots, K\}^N$ is the knowledge vector across N subjects, $e \in \{1, \ldots, E\}$ is energy, and $d \in \{1, \ldots, D\}$ is the current day. The **action** space is $a \in \{0, \ldots, N\}$: study subject i or rest ($a = N$). Dynamics:

$$\text{Forgetting:} \qquad k_i \leftarrow \max(0, \lfloor k_i \cdot e^{-\lambda_i} \rfloor) \quad \forall i \neq a \tag{1}$$

$$\text{Learning:} \qquad k_a \leftarrow \min(K, k_a + \lfloor \tfrac{e}{E} g_{\max}(1 + \text{bonus}(a)) \rfloor) \tag{2}$$

$$\text{Energy:} \qquad e \leftarrow \text{clip}(1, E,\ e - c_{\text{study}} + 1 + c_{\text{rest}} + \xi) \tag{3}$$

where $\lambda_i \in [0.05, 0.2]$ is the subject-specific forgetting rate, $\text{bonus}(a) = \alpha \min_{p \in \text{prereq}(a)} k_p / K$ when all prerequisites exceed threshold β (zero otherwise), and $\xi \sim \text{Uniform}\{-1, 0, +1\}$ is the energy noise. The **reward** is $r_{\text{step}} = -0.1$ per step plus an exam-day reward $r_{\text{exam}} = w_i \sigma(k_i - K/2)$ on

[1] Code and experiment scripts: `https://github.com/muhammadov-q/cvar-dqn-study-scheduling`

subject i's exam day, where w_i is its credit weight and σ is the sigmoid. The sigmoid maps zero knowledge to near-zero reward, $K/2$ knowledge to $0.5w_i$, and mastery to w_i.

Configurations. We study three configurations of increasing scale (Table 1). Forgetting rates λ_i were chosen so subject 1 (the prerequisite-dependent one) decays the fastest, requiring active scheduling pressure; exam weights w_i are asymmetric to create a non-trivial allocation problem. The prerequisite bonus uses $\alpha = 0.5$ and threshold $\beta = K/2$. The energy step uses $c_{\text{study}} = 1$ when studying and $c_{\text{rest}} = 1$ when resting, giving an expected drift of 0 when alternating; the noise ξ is what makes the schedule risky.

Table 1: Environment configurations used in all experiments. Vectors are listed in subject order $(S_0, S_1, \ldots)$. Prerequisites are edges $j \to i$ meaning subject i requires j. All configurations use $E = 5$, energy noise $\xi \sim \text{Uniform}\{-1, 0, +1\}$, and per-step cost -0.1.

Config	N	K	D	$g_{\max}$	Forgetting $\boldsymbol{\lambda}$	Prereqs	Exam days $\mathbf{d}$	Weights $\mathbf{w}$
Small	3	5	7	2	(0.05, 0.15, 0.10)	$0 \to 1$	(4, 6, 5)	(3, 4, 2)
Medium	4	8	10	2	(0.05, 0.15, 0.10, 0.08)	$0 \to 1$	(4, 7, 6, 9)	(3, 4, 2, 3)
Large	5	10	14	3	(0.05, 0.15, 0.10, 0.08, 0.12)	$0 \to 1$, $1 \to 2$	(5, 8, 11, 13, 14)	(3, 4, 5, 2, 3)

State-space sizes are $|\mathcal{S}| \approx 7.6$k (Small), 328k (Medium), and 11.3M (Large).

3 Methods

We compare eight agents from three families. All learning agents observe a min-max normalized state vector.

3.1 Heuristics (non-learning baselines)

These three rules require no training and serve as sanity checks: a useful learning agent should beat all of them.

- **Uniform.** Round-robin: on day d, study subject $(d-1) \bmod N$. State-independent.
- **Most-Urgent-First (MUF).** Study the subject whose exam is closest in the future (and has not yet been examined). Ties broken by lower k_i.
- **Lowest-Knowledge-First (LKF).** Study the subject with the lowest current k_i, ignoring the exam schedule.

3.2 Value Iteration

Value Iteration (VI) [Bellman, 1957, Sutton and Barto, 2018] computes the optimal state-value function V^* exactly by iterating the Bellman optimality operator until convergence:

$$V_{t+1}(s) \;=\; \max_{a \in \mathcal{A}} \sum_{s'} P(s' \mid s, a) \left[R(s, a, s') + \gamma V_t(s') \right], \tag{4}$$

with $\gamma = 0.99$ and stopping tolerance $\max_s |V_{t+1}(s) - V_t(s)| < 10^{-6}$. The optimal policy is $\pi^*(s) = \arg\max_a \mathbb{E}_{s'}[R + \gamma V^*(s')]$. VI is exact but requires enumerating all states; it is tractable only on Small (7.6k states) and gives an upper-bound reference on the mean return achievable in the MDP.

3.3 Tabular Q-Learning

Tabular Q-Learning [Watkins and Dayan, 1992, Sutton and Barto, 2018] is the canonical model-free off-policy TD-control method. It maintains $Q(s, a)$ as a lookup table and applies the update

$$Q(s,a) \;\leftarrow\; Q(s,a) + \alpha\big[r + \gamma \max_{a'} Q(s',a') - Q(s,a)\big], \tag{5}$$

with ϵ-greedy exploration. We use $\alpha = 0.1$, $\gamma = 0.99$, and ϵ decayed geometrically from 1.0 to 0.01 (decay 0.995). The state is hashed by its tuple representation. Q-Learning is feasible on Small and Medium; on Large the table grows to millions of cells and learning becomes too slow to converge in 5,000 episodes.

3.4 DQN (with Double DQN)

DQN [Mnih et al., 2015] approximates $Q(s, a)$ by a neural network Q_θ and stabilizes learning with two devices: (i) an experience-replay buffer D that decorrelates sequential samples and (ii) a periodically-copied target network $Q_{\bar\theta}$ that fixes the bootstrap target. The standard DQN loss on a minibatch $\mathcal{B} \subset D$ is

$$\mathcal{L}_{\text{DQN}}(\theta) = \mathbb{E}_{(s,a,r,s')\sim\mathcal{B}}\Big[\big(r + \gamma \max_{a'} Q_{\bar\theta}(s',a') - Q_\theta(s,a)\big)^2\Big].$$

We use the *Double DQN* variant [Van Hasselt et al., 2016] which decouples action selection from evaluation in the target, mitigating the overestimation bias of max:

$$y_{\text{DDQN}} \;=\; r + \gamma\, Q_{\bar\theta}\big(s', \arg\max_{a'} Q_\theta(s',a')\big). \tag{6}$$

The network is a two-layer MLP (128 units, ReLU). DQN is our strongest pure mean-maximizing baseline and scales to all three configurations.

3.5 CVaR-DQN

Conditional Value at Risk. For a return random variable Z and risk level $\alpha \in (0, 1]$, $\text{CVaR}_\alpha(Z)$ is the expected return conditional on the worst α-fraction of outcomes [Rockafellar and Uryasev, 2000, Chow et al., 2015]:

$$\text{CVaR}_\alpha(Z) \;=\; \mathbb{E}\big[\, Z \,\big|\, Z \le \text{VaR}_\alpha(Z)\,\big], \qquad \text{VaR}_\alpha(Z) = \inf\{z : \Pr(Z \le z) \ge \alpha\}. \tag{7}$$

CVaR_α recovers the mean as $\alpha \to 1$ and the worst-case as $\alpha \to 0$. We use $\alpha = 0.1$, so CVaR measures the average of the worst-10% returns. This is the standard coherent risk measure used in risk-sensitive finance and RL.

Quantile representation. Following QR-DQN [Dabney et al., 2018], we approximate the return distribution $Z(s, a)$ by $M = 32$ quantile values $\theta_i(s, a)$ anchored at midpoint locations $\hat\tau_i = (2i - 1)/(2M)$ for $i = 1, \ldots, M$. The CVaR is then estimated by averaging the smallest $\lceil \alpha M \rceil$ quantiles, and the risk-sensitive action is

$$a^* = \arg\max_a \text{CVaR}_\alpha(s,a), \quad \text{CVaR}_\alpha(s,a) = \frac{1}{\lceil \alpha M \rceil} \sum_{i=1}^{\lceil \alpha M \rceil} \theta_{(i)}(s,a), \tag{8}$$

where $\theta_{(i)}(s, a)$ is the i-th smallest quantile. With $\alpha = 0.1$ and $M = 32$, three quantiles enter the average.

Loss. The network is trained by minimizing the quantile Huber loss [Dabney et al., 2018]:

$$\mathcal{L}_{\mathrm{QR}}(\theta) \;=\; \frac{1}{M}\sum_{i=1}^{M}\mathbb{E}_j\big[\,\big|\hat{\tau}_i - \mathbf{1}\{\delta_{ij}<0\}\big|\cdot\mathcal{L}^{\kappa}_{\mathrm{Huber}}(\delta_{ij})\,\big]\,, \tag{9}$$

where $\delta_{ij} = y_j - \theta_i(s,a)$ is the TD-error between predicted quantile i and target quantile j, $\mathcal{L}^{\kappa}_{\mathrm{Huber}}$ is the Huber loss with threshold $\kappa = 1$, and the indicator weighting (asymmetric in the sign of δ) pushes each quantile to its assigned position in the distribution. The bootstrap target uses Double-DQN-style decoupling: the next action is selected on the online network using the CVaR criterion (Eq. 8), and the corresponding quantiles are read from the target network.

Summary. Where DQN maximizes the *average* outcome, CVaR-DQN maximizes the *worst-case* outcome. The only architectural change is the output head (M values per action instead of 1) and the action criterion; replay, target net, optimizer, and exploration are identical to DQN.

3.6 Subject-Constrained CVaR-DQN (pilot)

A pilot extension that treats exam scheduling as a Multi-Objective Distributional Constrained MDP [Altman, 1999, Achiam et al., 2017]. Let $Z_i(s,a)$ denote the return attributable to subject i and define a passing threshold $T_i = \frac{1}{2}w_i$, equal to the sigmoid-midpoint reward (i.e. reward earned at $k_i = K/2$). The ideal problem

$$\max_a \sum_i \mathbb{E}[Z_i(s,a)] \quad \text{s.t.} \quad \mathrm{CVaR}_\alpha(Z_i(s,a)) \geq T_i \;\; \forall i$$

is impractical to solve directly: when constraints conflict the feasible set is empty and Bellman targets become unstable. We therefore use the squared-violation Lagrangian relaxation

$$a^* = \arg\max_a \; \sum_i \mathbb{E}[Z_i(s,a)] \;-\; \lambda \sum_i \big[\max(0, T_i - \mathrm{CVaR}_\alpha(Z_i(s,a)))\big]^2, \tag{10}$$

with $\lambda = 5$. The quantile network outputs $N \times M$ values per action (one quantile distribution per subject); the quantile Huber loss is summed over subjects. Per-subject rewards r_i are the exam scores attributed to subject i (zero elsewhere).

4 Experiments and Results

Setup. All RL agents are trained for 5,000 episodes and evaluated over 1,000 episodes across 5 seeds (42, 123, 456, 789, 1024). Metrics: **Mean Return** (average total reward), $\mathbf{CVaR}_{10\%}$ (mean of worst 10% returns), **Mean Worst Exam** (lowest individual exam score per episode).

4.1 Main Results

Table 2: Performance comparison across all configurations (mean over 5 seeds). Bold marks the best agent within each config for $\text{CVaR}_{10\%}$.

Config	Agent	Mean Return	$\text{CVaR}_{10\%}$	Worst Exam
Small	Uniform	0.68	0.03	0.20
	MUF	3.56	2.55	0.51
	LKF	0.76	0.22	0.15
	Value Iteration	5.00	3.44	0.15
	Q-Learning	2.85	1.80	0.27
	DQN	**4.97**	3.37	0.15
	CVaR-DQN	4.85	**3.48**	0.15
	SC-CVaR-DQN	4.68	3.23	0.22
Medium	Uniform	−0.71	−0.78	0.04
	MUF	3.11	1.38	0.24
	LKF	−0.48	−0.72	0.04
	Q-Learning	1.47	0.59	0.21
	DQN	**4.86**	2.86	0.04
	CVaR-DQN	4.50	**3.19**	0.04
	SC-CVaR-DQN	4.53	2.40	0.04
Large	Uniform	−1.05	−1.29	0.02
	MUF	8.50	2.25	0.14
	LKF	−0.88	−1.25	0.02
	DQN	**11.47**	8.25	0.01
	CVaR-DQN	10.62	**8.27**	0.01
	SC-CVaR-DQN	11.19	8.10	0.01

Key findings. DQN achieves the highest mean return across all configurations, closely matching Value Iteration on Small (4.97 vs. 5.00). **CVaR-DQN consistently achieves higher $\text{CVaR}_{10\%}$**, confirming more robust schedules: on Medium, DQN gets 4.86 mean vs. CVaR-DQN's 4.50 (−7% mean) but $\text{CVaR}_{10\%}$ flips to 2.86 vs. 3.19 (+12% worst-case)—the mean-risk trade-off the algorithm is designed to make. SC-CVaR-DQN is competitive on mean return but underperforms on aggregate CVaR; we analyze its specific behavior in §4.4. Among heuristics, Most-Urgent-First performs best. All three deep RL agents converge within ∼2,000 episodes (Appendix Figure 2); the full per-agent comparison for every configuration is in Appendix Figure 1.

4.2 Ablation Study

We ablate one component at a time on the medium configuration (Table 3, visual in appendix Figure 5).

Table 3: Ablation results (medium config, mean over 5 seeds). Bold values mark the better agent within each row for $\text{CVaR}_{10\%}$.

Ablation	Mean Return		$\text{CVaR}_{10\%}$	
	DQN	CVaR-DQN	DQN	CVaR-DQN
Baseline	**4.79**	4.50	2.91	**3.19**
No forgetting	5.63	5.59	4.34	**4.95**
Constant energy	7.01	7.01	7.01	7.01
No prerequisites	**4.86**	4.58	2.91	**3.17**
Equal weights	0.82	0.74	**0.33**	0.31
$\lambda = 0.05$	**4.85**	4.52	2.88	**3.13**
$\lambda = 0.2$	4.83	4.54	2.96	**3.14**
$\lambda = 0.4$	4.57	4.36	**3.03**	2.86

Constant energy eliminates all variance: both agents achieve the deterministic optimum (7.01)—when energy noise is removed there is no risk to be sensitive to, confirming energy stochasticity is the sole source of uncertainty differentiating the two. **No forgetting** boosts both agents and grows CVaR-DQN's advantage (4.95 vs. 4.34). **High forgetting** ($\lambda = 0.4$) is the only condition where DQN's CVaR exceeds CVaR-DQN's—rapid decay rewards aggressive late study over caution. **Prerequisites** have minimal effect.

4.3 Robustness Tests

We evaluate agents trained on normal conditions under three perturbations (Table 4, appendix Figure 6).

Table 4: Robustness results (mean over 5 seeds). Bold values mark the best agent within each row for $\text{CVaR}_{10\%}$.

Test	Mean Return			$\text{CVaR}_{10\%}$		
	DQN	CVaR-DQN	SC-CVaR-DQN	DQN	CVaR-DQN	SC-CVaR-DQN
Normal	**4.86**	4.50	4.53	2.90	**3.19**	2.40
Wide energy noise	**4.08**	3.79	3.69	**1.56**	1.55	1.13
Compressed exams	2.76	**2.92**	1.89	0.80	**1.34**	0.30
High variance	**4.07**	3.56	3.52	**1.54**	1.27	1.09

Under **compressed exams**, CVaR-DQN maintains the highest $\text{CVaR}_{10\%}$ (1.34 vs. 0.80 for DQN)—its conservative strategy transfers naturally when exam clustering removes recovery slack. Under **wide energy noise** and **high variance** (doubled forgetting + wide noise), all three RL agents degrade; DQN retains slightly higher CVaR. **SC-CVaR-DQN does worse than both baselines under all perturbations**—the constraint thresholds T_i are tuned to the training distribution, so the agent over-hedges against the wrong type of risk under shift.

4.4 Pilot: Subject-Constrained CVaR-DQN

We examine per-subject failures: Table 5 reports the fraction of episodes in which each subject's exam score falls below $T_i = 0.5w_i$.

Table 5: Per-subject failure rate (fraction of episodes with exam score $< 0.5w_i$, mean over 5 seeds). S_i = subject i; bold marks the lowest failure rate per subject among the three RL agents. "—" indicates the subject does not exist in this configuration.

Config	Agent	S_0	S_1	S_2	S_3	S_4
Small	Value Iteration	0.00	0.08	1.00	—	—
	DQN	**0.00**	**0.10**	1.00	—	—
	CVaR-DQN	0.11	0.11	1.00	—	—
	SC-CVaR-DQN	0.02	0.16	1.00	—	—
Medium	DQN	**0.04**	0.30	1.00	**0.50**	—
	CVaR-DQN	0.40	0.46	1.00	**0.14**	—
	SC-CVaR-DQN	**0.04**	**0.08**	1.00	0.53	—
Large	DQN	**0.00**	**0.01**	0.04	1.00	0.12
	CVaR-DQN	0.20	0.20	**0.01**	1.00	**0.04**
	SC-CVaR-DQN	**0.00**	0.05	0.02	1.00	0.13

Diagnostic. On every configuration, at least one subject fails 100% of the time—and crucially, this holds *even for Value Iteration*, the DP optimum. The lowest-weight subject ($w = 2$: Subject 2 on Small/Medium, Subject 3 on Large) has maximum exam reward ≈ 1.85, which is smaller than the marginal reward of spending those days on higher-weight subjects; the reward-optimal policy rationally sacrifices it.

What SC-CVaR-DQN achieves. The one positive: **on Medium, SC-CVaR-DQN reduces Physics failure (S_1) from 30% (DQN) and 46% (CVaR-DQN) to 8%**—a 4–6× reduction—while Math failure stays low (0.04). This is the algorithm working as intended: the per-subject distributional forecast and Lagrangian penalty reallocate effort toward an at-risk-but-protectable subject.

What it does *not* achieve. (i) Cannot protect environment-infeasible subjects: increasing λ from 5 to 100 makes no difference. (ii) Underperforms on aggregate metrics: mean return below DQN's by ~ 0.3 on every config; $\text{CVaR}_{10\%}$ below both DQN and CVaR-DQN on Medium (2.40 vs 2.86 vs 3.19). (iii) Degrades under distribution shift (Table 4)—thresholds tuned to the training distribution misfire when test conditions change. On a deliberately easy "Fair" configuration (equal weights, no forgetting, no prerequisites), DQN/CVaR-DQN reach 0% any-failure while SC-CVaR-DQN sits at 2–3%—the framework permits no-failing, and the algorithm is a small cost when no constraint actually binds.

5 Conclusion

We presented a risk-sensitive RL approach to exam study scheduling. CVaR-DQN consistently produces higher $\text{CVaR}_{10\%}$ than DQN across all three configurations, at a small cost in mean return—the mean-risk trade-off in finance. Ablations show that energy stochasticity is the dominant source of outcome variance (removing it makes DQN and CVaR-DQN identical), while forgetting affects difficulty level uniformly. The SC-CVaR-DQN pilot validates the formulation (per-subject distributional constraints are computable and trainable) and yields a clear positive (4–6× reduction in Physics failure on Medium) embedded in clear negatives (lower aggregate metrics, worse robustness, no help on environment-infeasible subjects). The root cause of the negatives is fixed thresholds—they neither adapt to environment dynamics nor to test-time distribution—and adaptive thresholding is the natural follow-up direction.

Limitations & future work. The environment uses discrete knowledge levels and a uniform energy-noise model; real cognitive energy is autocorrelated and influenced by exogenous factors. A natural next step is to fit the energy and forgetting processes to *real student data* (sleep-tracker logs, learning-platform telemetry) so the agent can be trained on empirically grounded dynamics. SC-CVaR-DQN training is $\sim$1.3$\times$ slower than CVaR-DQN due to the N-fold quantile output.

Acknowledgments

This work made use of several AI assistants. The implementation of CVaR-DQN, SC-CVaR-DQN, and the supporting experiment harness was written by the author with the assistance of a large language model (Anthropic Claude); the CVaR-DQN implementation follows Dabney et al. [2018] and the SC-CVaR-DQN formulation is original to this work. Perplexity AI was used to locate and cross-reference background literature. Google Gemini was used for English-language polishing, grammar, and word choice in the report text. All algorithmic choices, hyperparameters, experimental design, results interpretation, and the substantive content of this report are the author's own.

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A Training Parameters

All experiments use seeds $\mathcal{Z} = \{42, 123, 456, 789, 1024\}$ (one seed per training+evaluation run). Each agent is trained for 5,000 episodes and evaluated for 1,000 episodes; reported numbers are means over the five seeds (bootstrap 95% CIs computed in plots). The seed controls Python's `random`, NumPy, PyTorch, and the environment's internal RNG simultaneously.

Table 6: Shared training schedule. Identical for all RL agents on all configurations.

Parameter	Value
Random seeds	$\{42, 123, 456, 789, 1024\}$
Training episodes	5,000
Evaluation episodes	1,000
Discount factor γ	0.99
ϵ-greedy schedule	$\epsilon_0 = 1.0 \to \epsilon_\infty = 0.01$, geometric decay 0.995/episode

Table 7: Agent-specific hyperparameters. The DQN, CVaR-DQN, and SC-CVaR-DQN agents share the same optimization/replay setup; risk-related parameters differ.

Parameter	Q-Learning	DQN	CVaR-DQN	SC-CVaR-DQN
Learning rate	0.1	10^{-3}	10^{-3}	10^{-3}
Optimizer	–	Adam	Adam	Adam
Hidden architecture	tabular	MLP(128,128)	MLP(128,128)	MLP(128,128)
Activation	–	ReLU	ReLU	ReLU
Replay buffer size	–	50,000	50,000	50,000
Minibatch size	–	64	64	64
Target-network update freq.	–	100 steps	100 steps	100 steps
Double-DQN target	–	yes	yes (CVaR-selection)	yes (constrained-selection)
Gradient clip (ℓ_2 max-norm)	–	10	10	10
Quantiles M	–	–	32	32
Risk level α	–	–	0.10	0.10
Huber threshold κ	–	–	1.0	1.0
Subject thresholds T_i	–	–	–	$0.5\, w_i$
Lagrangian multiplier λ	–	–	–	5.0

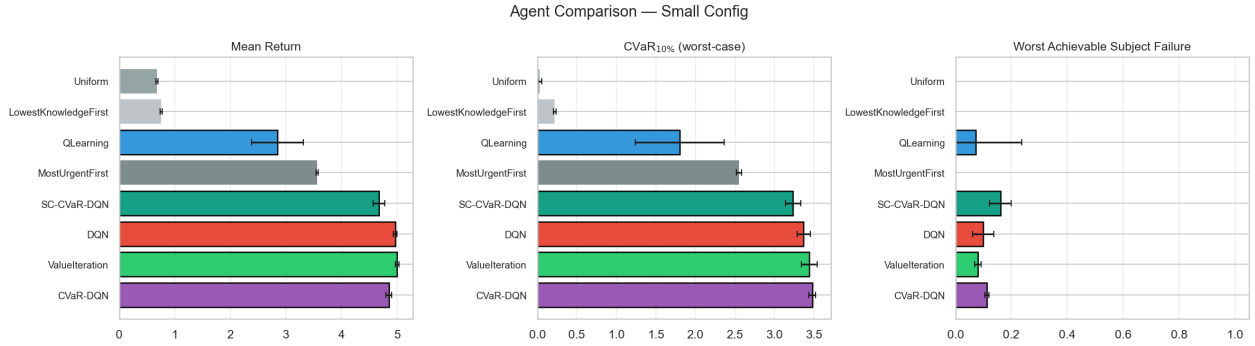
Table 8: Heuristic and planning agents. No learning rates or replay; these depend only on the configuration.

Agent	Configuration
Uniform	State-independent round-robin: action $= (d-1) \bmod N$.
Most-Urgent-First	Pick the not-yet-examined subject with the earliest exam day.
Lowest-Knowledge-First	Pick the subject with the lowest current knowledge k_i.
Value Iteration	$\gamma = 0.99$; sweep convergence tolerance 10^{-6}. Small only (7.6k states).

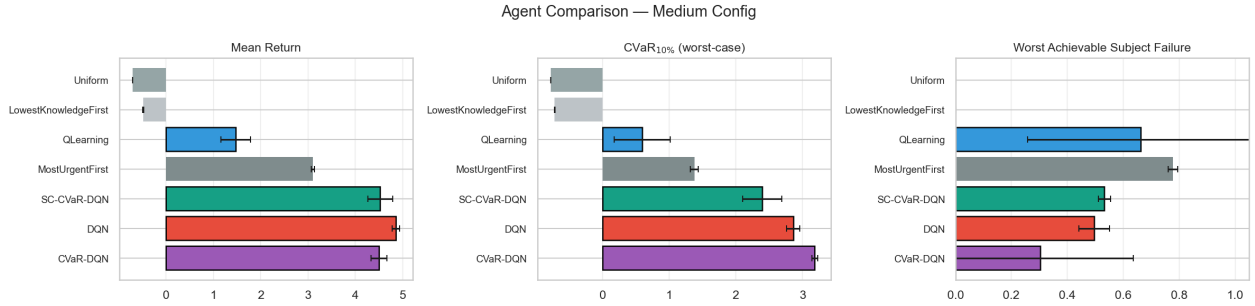
The reported main results (Table 2) run all eight agents across all three configurations for which they are tractable. Ablation (Table 3) and robustness (Table 4) studies use the Medium configuration with the same five seeds and the same training schedule.

Software & reproducibility. Python 3.10, PyTorch 2.x, NumPy, Gymnasium. The full codebase including the environment, agents, experiment scripts, configurations, and plotting pipeline is available at `https://github.com/muhammadov-q/cvar-dqn-study-scheduling`.

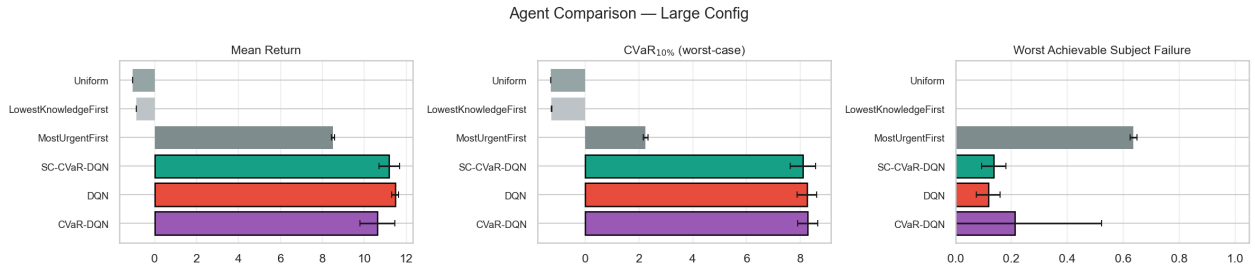
B Supplementary Figures



(a) Small configuration

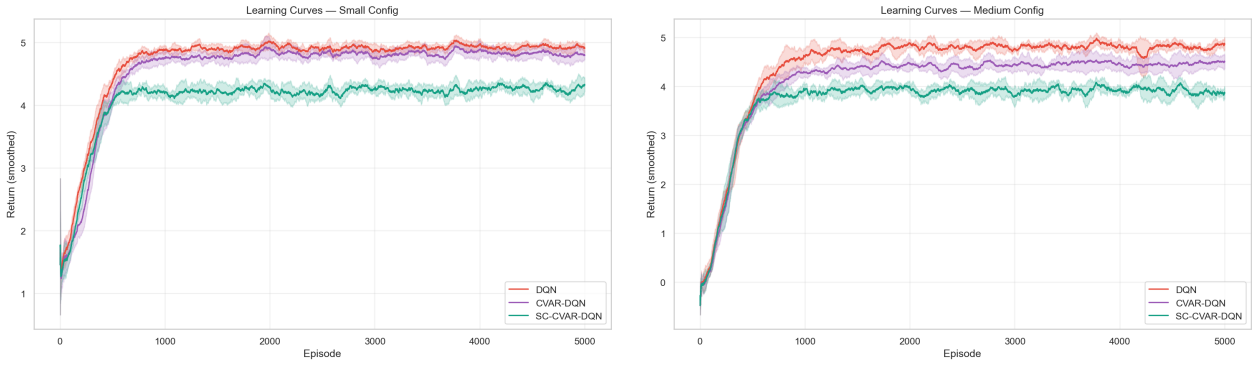


(b) Medium configuration



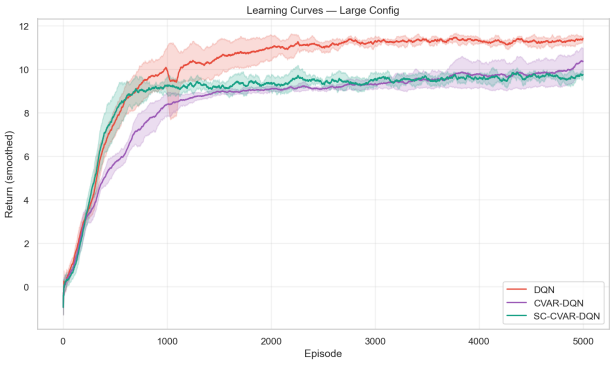
(c) Large configuration

Figure 1: Per-agent comparison across all three configurations. Bars sorted by $\text{CVaR}_{10\%}$; "worst achievable subject failure" excludes universally-infeasible subjects (failure 1.0 even for Value Iteration).



(a) Small

(b) Medium



(c) Large

Figure 2: Learning curves (mean ± std over 5 seeds) for DQN, CVaR-DQN, and SC-CVaR-DQN.

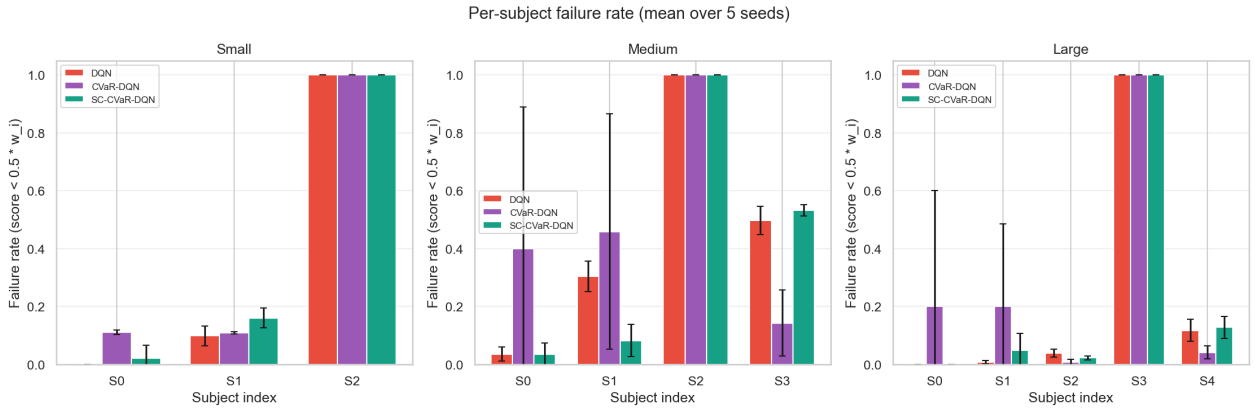


Figure 3: Per-subject failure rate (mean ± std over 5 seeds), DQN vs. CVaR-DQN vs. SC-CVaR-DQN. Same data as Table 5.

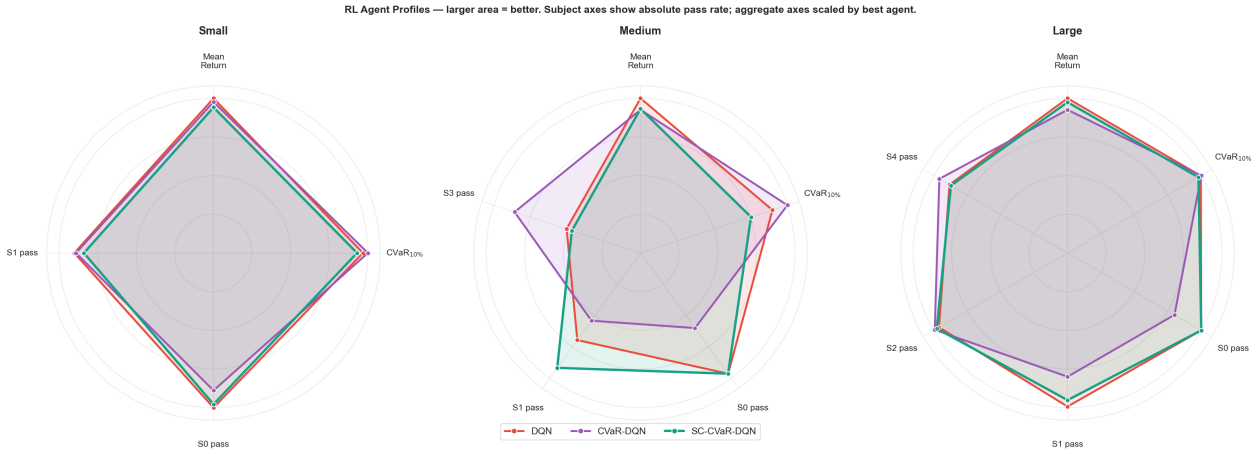


Figure 4: Multi-dimensional profile of each RL agent per configuration. Subject axes show absolute pass rate (1− failure rate); aggregate axes (Mean Return, $\text{CVaR}_{10\%}$) are scaled so the best agent on each axis sits at 1. Universally-infeasible subjects are omitted. Larger enclosed area = better.

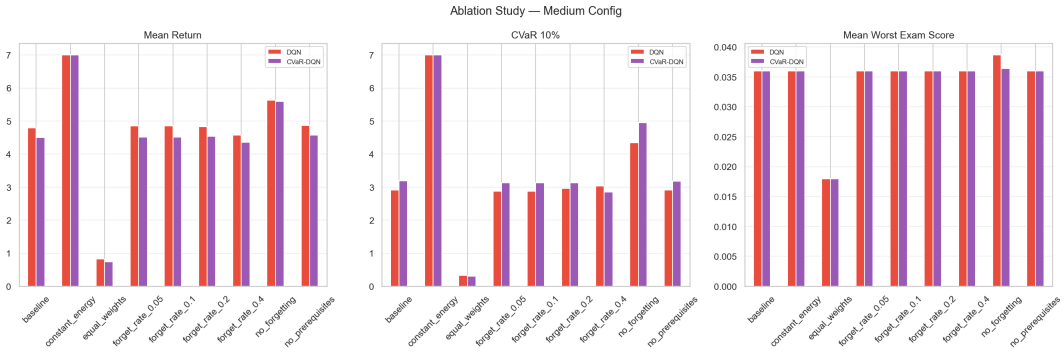


Figure 5: Ablation study visual summary; same data as Table 3.

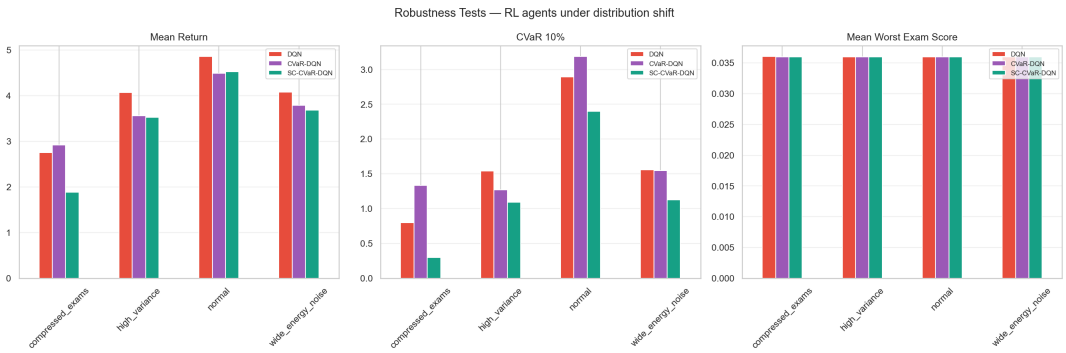


Figure 6: Robustness tests visual summary; same data as Table 4.